BAIO Learning System

I PUC BIOLOGY

MID-TERM PREPARATION

Score High in 2026

- Important Topics
- Diagrams to Practise
- Exam Tips



Narendra Sir
Biology Lecturer
Founder/CEO, BAIO Learning System

How to use this video

- Pause on every topic and **write the answer** in your notebook
- Then check it with the slide and correct mistakes
- Draw every diagram **as given by Narendra Sir**
- Revise the key facts again just before the exam

CHAPTER

1

The Living World

📝 4 marks in the paper · 3.8% weightage

Narendra Sir · Founder/CEO, BAIO

Key facts for MCQs

Part A · 1 mark each · 2 Qs

- **Species** is the basic unit of classification
- Scientific name: genus with a capital letter, species with a small letter, in italics or underlined — *Mangifera indica* (mango)



NARENDRA SIR SAYS

What to study: Species as the basic unit and the rules for writing scientific names

How to study: Write five scientific names every day: genus with a capital letter, species small, both underlined

Binomial nomenclature

Part B · 2 marks

- Naming an organism with **two words**: generic name + specific epithet
- Example: *Homo sapiens*
- Proposed by **Carolus Linnaeus**

CHAPTER

2

Biological Classification

📝 10 marks in the paper · 9.5% weightage

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Key facts for MCQs

Part A · 1 mark each · 2 Qs

- Fungal cell wall is made of **chitin**
- Red rot of sugarcane is caused by a **fungus**, *Colletotrichum falcatum*



NARENDRA SIR SAYS

What to study: Fungal cell wall, fungal diseases and the features of each kingdom

How to study: Make a five-kingdom chart with one example each; learn every disease with its causative organism

Kingdom Monera

Part C · 3 marks

- Prokaryotic, unicellular; no membrane-bound organelles
- Cell wall of **peptidoglycan**
- Most abundant; found even in hot springs, deserts, snow, deep oceans
- Autotrophic (photo / chemosynthetic) or heterotrophic
- Reproduce mainly by fission
- Examples: *E. coli*, *Nostoc*, *Anabaena*, *Mycoplasma*

Whittaker's five kingdoms (1969)

Part D · 5 marks

Kingdom	Key features	Examples
Monera	Prokaryotic, unicellular, peptidoglycan wall	Bacteria, cyanobacteria
Protista	Unicellular eukaryotes, aquatic	*Chlamydomonas*, *Paramecium*
Fungi	Chitin wall, heterotrophic	*Mucor*, *Agaricus*, yeast
Plantae	Cellulose wall, autotrophic	Algae to flowering plants
Animalia	No cell wall, holozoic	Sponges to mammals

Criteria: cell structure, body organisation, nutrition, reproduction, phylogeny

CHAPTER 2 · 10 MARKS

Five kingdoms — merits and demerits

Part D · 5 marks

Merits

- Prokaryotes separated from eukaryotes
- Unicellular eukaryotes grouped in Protista
- Fungi separated from plants
- Reflects phylogeny better than two kingdoms

Demerits

- Protista is a mixed group (*Chlamydomonas* with *Paramecium*)
- Archaebacteria and eubacteria together in Monera
- Viruses, viroids and lichens not placed
- Photosynthetic cyanobacteria kept away from plants

CHAPTER

3

Plant Kingdom

📝 10 marks in the paper · 9.5% weightage

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Key facts for MCQs

Part A · 1 mark each · 3 Qs

- Bryophytes = **amphibians of the plant kingdom**
- *Cycas* is a **living fossil** among gymnosperms
- *Chlorella*: protein-rich alga used by space travellers



NARENDRA SIR SAYS

What to study: Bryophytes, gymnosperms and the economic importance of algae

How to study: Learn the algae table (pigments, stored food, cell wall) and say it aloud once a day

Economic importance of algae

Part B · 2 marks

- Food: *Porphyra*, *Laminaria*, *Sargassum*; supplements *Chlorella*, *Spirulina*
- Algin (brown algae) and carrageen (red algae) used commercially
- Agar from *Gelidium* and *Gracilaria* — for culture media, ice-creams, jellies
- Algae fix about half of the total CO_2 on earth

Classes of algae

Part D · 5 marks

Class	Common name	Pigments	Stored food	Cell wall
Chlorophyceae	Green algae	Chl a, b	Starch	Cellulose, pectose
Phaeophyceae	Brown algae	Chl a, c, fucoxanthin	Mannitol, laminarin	Cellulose, algin
Rhodophyceae	Red algae	Chl a, d, phycoerythrin	Floridean starch	Cellulose, pectin, polysulphate esters

CHAPTER

4

Animal Kingdom

📝 16 marks in the paper · 15.2% weightage

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Key facts for MCQs

Part A · 1 mark each · 3 Qs

- **Water vascular system** — Echinodermata
- Mammals are **homoiothermous** (warm-blooded)
- Sponges belong to phylum **Porifera**



NARENDRA SIR SAYS

What to study: The one unique feature of each phylum: water vascular system, pores, warm blood

How to study: Make flashcards: feature on one side, phylum on the other, and practise both ways

Chordates and non-chordates

Part C · 3 marks

Chordates	Non-chordates
Notochord present	Notochord absent
CNS dorsal, hollow, single	CNS ventral, solid, double
Pharyngeal gill slits present	Gill slits absent
Heart ventral	Heart dorsal (if present)
Post-anal tail present	Post-anal tail absent

Phylum Arthropoda

Part D · 5 marks

- Largest phylum — about two-thirds of all named species
- Bilateral, triploblastic, segmented, coelomate; chitinous exoskeleton
- **Jointed appendages**; body = head, thorax, abdomen
- Respiration: gills, book gills, book lungs, tracheae; **open circulation**
- Excretion by **Malpighian tubules**; antennae, compound eyes, statocysts
- Examples: *Apis*, *Bombyx*, *Laccifer*, *Anopheles*, *Locusta*, *Limulus*

Identify the phylum

Part D · 5 marks

Key characters	Phylum
Proboscis gland, collar and trunk	Hemichordata
Comb plates, bioluminescence	Ctenophora
Jointed appendages, Malpighian tubules	Arthropoda
Notochord, hollow nerve cord, gill slits	Chordata
Calcareous shell, head-foot-visceral mass, radula	Mollusca

CHAPTER

5

Morphology of Flowering Plants

📝 11 marks in the paper · 10.5% weightage

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Key facts for MCQs

Part A · 1 mark each · 1 Q

- Edible part of mango: fleshy **mesocarp**
- Venation = arrangement of veins in the lamina Part B · 2 marks
- **Reticulate** — network of veinlets (dicots)
- **Parallel** — veins run parallel (monocots)



NARENDRA SIR SAYS

What to study: Parts of fruits and types of venation with examples

How to study: Pick up a real leaf and a mango at home and name each part

Phyllotaxy

Part C · 3 marks

- Arrangement of leaves on the stem
- **Alternate** — one leaf per node: china rose, mustard, sunflower
- **Opposite** — a pair per node: *Calotropis*, guava
- **Whorled** — more than two per node: *Alstonia*

Placentation

Part D · 5 marks

- Arrangement of ovules within the ovary
- **Marginal** — pea
- **Axile** — china rose, tomato, lemon
- **Parietal** — mustard, *Argemone* (false septum)
- **Free central** — *Dianthus*, *Primrose*
- **Basal** — sunflower, marigold

CHAPTER

6

Anatomy of Flowering Plants

📝 5 marks in the paper · 4.8% weightage

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Key facts for MCQs

Part A · 1 mark each · 2 Qs

- Root hairs are outgrowths of **epidermal cells**
- Stomata are regulated by bean-shaped **guard cells**



NARENDRA SIR SAYS

What to study: Epidermal tissue: root hairs, stomata and guard cells

How to study: Draw a stoma and label it; compare bean-shaped (dicot) and dumb-bell (grass) guard cells

Open and closed vascular bundles

Part C · 3 marks

Open	Closed
Cambium present	Cambium absent
Secondary growth possible	No secondary growth
Dicot stems	Monocot stems
Bundle sheath usually absent	Sclerenchymatous bundle sheath

CHAPTER

7

Structural Organisation in Animals (Frog)

📝 4 marks in the paper · 3.8% weightage

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Key facts for MCQs

Part A · 1 mark each · 1 Q

- Frogs are **cold-blooded (poikilotherms)** — body temperature changes with the surroundings
- To escape extreme heat and cold, frogs shelter in deep burrows
- **Aestivation** — summer sleep of the frog
- **Hibernation** — winter sleep of the frog



NARENDRA SIR SAYS

What to study: Cold-blooded frog, hibernation and aestivation

How to study: Remember: hibernation = winter, aestivation = summer; then draw the digestive system twice

Digestive system of frog

Part C · 3 marks

- Alimentary canal is **short** — frogs are carnivores
- Mouth → buccal cavity → pharynx → oesophagus → stomach → intestine → rectum → **cloaca**
- Food caught by the **bilobed tongue**; stomach digests with **HCl and gastric juice**
- **Bile** (liver, stored in gall bladder) and **pancreatic juice** reach the duodenum by the common bile duct
- Absorbed by **villi and microvilli**; waste leaves through the cloaca

Diagram — Digestive system of frog

Part C · 3 marks

✏ Draw the diagram as given by Narendra Sir

Labels to include: Mouth · buccal cavity · tongue · pharynx · oesophagus · stomach · duodenum · intestine (ileum) · rectum · cloaca · cloacal aperture · liver · gall bladder · common bile duct · pancreas

CHAPTER

8

Cell: The Unit of Life

📝 21 marks in the paper · 20% weightage

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Key facts for MCQs

Part A · 1 mark each · 2 Qs

- Plant cell wall is made of **cellulose**
- Ribosomes are made in the **nucleolus**
- **Mesosomes**: infoldings of bacterial plasma membrane — help in wall formation, DNA replication, respiration, secretion Part B · 2 marks



NARENDRA SIR SAYS

What to study: Cell-wall material, the nucleolus, mesosomes and what each organelle does

How to study: Write a two-column list "structure → organelle" and test yourself until it is quick

Plant cell and animal cell

Part B · 2 marks

Plant cell	Animal cell
Cellulose cell wall	No cell wall
Plastids present	No plastids
One large central vacuole	Vacuoles small or absent
Centrioles usually absent	Centrioles present

Diagram — Plant cell

Diagram · draw with labels

✏ Draw the diagram as given by Narendra Sir

Labels to include: Cell wall · plasma membrane · cytoplasm · nucleus · nucleolus · chloroplast · mitochondrion · central vacuole · tonoplast · Golgi apparatus · ER · ribosomes · plasmodesmata

Fluid mosaic model

Part D · 5 marks

- Proposed by **Singer and Nicolson (1972)**
- **Lipid bilayer**: polar heads outside, non-polar tails inside
- **Quasi-fluid** — lipids and proteins move sideways
- Proteins: **peripheral** (surface) and **integral** (buried)
- RBC membrane: about 52% protein, 40% lipid
- Selectively permeable — passive and active transport

Mitochondrion

Part D · 5 marks

- Double membrane-bound; inner membrane folds into **cristae**
- Intermembrane space and **matrix**
- **F_0–F_1 particles** on cristae make ATP
- Matrix: circular DNA, RNA, 70S ribosomes, Krebs cycle enzymes
- Site of aerobic respiration; semi-autonomous

Diagram — Mitochondrion

Part D · 5 marks

✏ Draw the diagram as given by Narendra Sir

Labels to include: Outer membrane · inner membrane · cristae · intermembrane space · matrix · F_o–F_1 particles · circular DNA · 70S ribosomes

Structures and their organelles

Part D · 5 marks

Structure	Found in
Cisternae	Golgi apparatus
Cristae	Mitochondrion
Thylakoids	Chloroplast
Nucleolus	Nucleus
Tonoplast	Vacuole

CHAPTER

9

Biomolecules

📝 12 marks in the paper · 11.4% weightage

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Key facts for MCQs

Part A · 1 mark each · 2 Qs

- **Purines:** adenine, guanine · **Pyrimidines:** cytosine, thymine, uracil
- Proteins are polymers of **amino acids**
- **Polysaccharides**: chains of monosaccharides joined by glycosidic bonds — cellulose, starch, glycogen, inulin, chitin Part B · 2 marks



NARENDRA SIR SAYS

What to study: Purines and pyrimidines, and the building blocks of proteins and polysaccharides

How to study: Trick: "Pure As Gold" — purines are Adenine and Guanine; write every monomer → polymer pair

Structure of a nucleotide

Part C · 3 marks

- **Nitrogenous base** + **pentose sugar** + **phosphate**
- Base + sugar (N-glycosidic bond at 1′C) = **nucleoside**
- Nucleoside + phosphate (phosphoester bond at 5′C) = **nucleotide**
- Nucleotides join by 3′–5′ **phosphodiester bonds**

Factors affecting enzyme activity

Part D · 5 marks

- **Temperature** — optimum; low = inactive (reversible), high = denatured
- **pH** — each enzyme has an optimum pH
- **Substrate concentration** — velocity rises, then reaches V_{max}; K_m = [S] at ½ V_{max}
- **Activators and inhibitors** — malonate competitively inhibits succinic dehydrogenase

CHAPTER

10

Cell Cycle and Cell Division

📝 12 marks in the paper · 11.4% weightage

Narendra Sir · Founder/CEO, BAIO

Key facts for MCQs

Part A · 1 mark each · 2 Qs

- **Metaphase**: chromosomes at the equator, spindle fibres on kinetochores
- **Crossing over**: exchange between non-sister chromatids, by recombinase

  Part B · 2 marks
- Crossing over occurs in **pachytene** of Prophase I



NARENDRA SIR SAYS

What to study: Stages of mitosis and the five stages of Prophase I

How to study: Trick: "Lazy Zebras Prefer Dancing Daily" = leptotene, zygotene, pachytene, diplotene, diakinesis

Mitosis and meiosis

Part C · 3 marks

Mitosis	Meiosis
Somatic cells	Reproductive cells
One division, 2 cells	Two divisions, 4 cells
2n → 2n	2n → n
No crossing over	Crossing over in Prophase I
Identical daughter cells	Genetically different cells

Phases of the cell cycle

Part D · 5 marks

- **G_1** — cell grows; no DNA replication
- **S** — DNA replicates (2C → 4C)
- **G_2** — proteins made for mitosis
- **M phase** — karyokinesis (P, M, A, T) + cytokinesis
- **G_0** — quiescent stage; cells exit from G_1

Diagram — Cell cycle

Diagram · draw with labels

✏ Draw the diagram as given by Narendra Sir

Labels to include: G_1 · S · G_2 · M phase · G_0 · interphase · direction arrows

Diagrams to practise

1 Plant cell

2 Mitochondrion

3 Cell cycle

4 Digestive system of frog

Narendra Sir's tips — how to study

- Read a topic, close the book, write it from memory
- Practise every diagram at least three times
- Learn examples and scientific names with each topic
- Revise differences as tables
- Use the **BAIO Learning System** app daily for MCQs and flashcards

Narendra Sir's tips — in the exam hall

- Read the whole paper first; plan your choices
- Part A: only the **first answer** is evaluated
- Match the marks: 2 marks = 2 sentences, 5 marks = 200–250 words
- Write in points, underline key words, give examples
- **Unlabelled diagrams get no marks**
- Keep the last 10 minutes to check



All the best!

Work hard, stay confident, score high.

Narendra Sir

Biology Lecturer · Founder/CEO, BAIO Learning System

Practise more on the BAIO Learning System app